

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
Student Name:	S. Jagadeesh	Reg. No.:	192421406
Section / Batch:	B.Tech - IT	Date:	24/07/26

Code of Conduct

I S. Jagadeesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

80

100

100/100

spectrum auction 45 MHz, 25 kHz for every simplex channel.
45,000 kHz

single = $25 \times 2 = 50$ kHz
channel

$$\text{duplex} = \frac{45,000}{50} = 900$$

② carriers runs in 50, 7-cell frequency

$$\begin{aligned} \text{single} &\Rightarrow \frac{\text{Total channels}}{\text{cell frequency}} \\ &= \frac{900}{7} = 80 \end{aligned}$$

③ freq spacing of 200 kHz, 80 is out of carriers, each carrying 8 time slots

$$\text{voice channels} = 100 \times 8 = 800$$

④ base station radiates 80W, handset transmits 4W

$$10 \log_{10}(80) \Rightarrow 19.0$$

$$\Rightarrow 13 \text{ dB}$$

⑤ -60 dBm serving signal, co-channel interference at -92 dBm

$$\begin{aligned} \text{carrier-to-interference} &= (\text{serving signal}) - (\text{co-channel interference}) \\ &= -60 - (-92) \\ &= -60 + 92 \\ &\Rightarrow 32 \end{aligned}$$

⑥ operator holds 42 MHz, assigning 300 kHz per channel, 7 cell pattern

$$\text{single cell} = \frac{140}{7} = 20$$

$$\frac{42,000}{300} = 140$$

⑦ 64W of power to a receiver, path loss exponent 4

$$\Rightarrow 2^4 = 16$$

$$\text{transmit power} = \frac{64}{16} = 4$$

⑧ planer to double cell's radius, SNR at the new edge

$\Rightarrow$ inverse square power law

$$\Rightarrow 1/3 \text{ dB}$$

⑨ cluster size is 7, 15 voice channels

$$\text{total channel} = \text{voice channel} \times \text{cluster size} = 15 \times 4 = 60$$

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total channel = voice channel $\times$ cluster size = $15 \times 4 = 60$

34 7 cell cluster, 900 simultaneous users

$$\text{channels} = \frac{900}{7} = 128.57 \approx 128$$

35 A TDMA frame spans 4.8 ms, 8 equal slots

$$\text{individual} = \frac{4.8}{8} = 0.6$$

36 at 100 km/h, at 1.8 GHz, Doppler shift

$$100 \text{ km/h} = 41.6 \text{ m/s}, \quad \Delta f = v \times \frac{f}{c}$$

$$\text{Doppler} = v \times \frac{f}{c} = \frac{41.6 \times 1.8 \times 10^9}{3 \times 10^8} = 2.496 \times 10^2$$

37 2,50,000 simultaneous calls, 70,000 per minute, 4 minutes

$$\text{afterload} = \frac{2,50,000}{70,000} = 3.57 \approx 4$$

38 hexagonal layout 9 cells, cell has 600 users

$$\text{cluster capacity} = 600 \times 9 = 5400$$

39 context process 60,000 per sec, 0.4% then dropped

$$\text{dropped every second} = 60,000 \times 0.4 = 240$$

40 A base station in 1800 MHz, network at 43 dBm, 2km-radius cell, the signal -87 dBm

$$\Rightarrow 2 \times 2 = 4, \quad \frac{1800}{4} = 450, \quad \frac{450}{4} = 112.5$$

$$\Rightarrow -(10.4) - (-87)$$

$$= -97.4$$

41 base station covers 2.5 km², 150 identical stations

$$\text{geographic area} = 150 \times 2.5 = 375$$

42 frequency reuse 6, 48 MHz spectrum allocation, each channel 30 kHz

$$\Rightarrow \text{Total channel} = 6 \times 48 \times 200$$

$$= 57,600$$

6000 users per cell, 150,000 base total users

$$\text{Paging cycle} = \frac{150000}{6000} = 25$$

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all calls for a min, every 3 minutes

$$\text{Handoffs} = \frac{9}{3} = 3$$

$$-85 \text{ dBm, noise floor } -115$$

$$\Rightarrow \text{sig to noise} = - (85) - (-115) \\ = \cancel{30} 30 \text{ dB}$$

46) A cell can serving 250 users, 150 kbps

$$\Rightarrow 250 \times 150 = 37,500$$

47) 15000 subscribers, 4 call per hour, 2.5 minute

$$\Rightarrow 15000 \times 4 \times 2.5 = 15,000$$

$$= \frac{15000}{60} = 250$$

48) Total bandwidth = 250 x 12 = 3000

$$\Rightarrow 3 \text{ MHz}$$

49) 1500 simultaneous connections, 600 kbps on average

$$\Rightarrow 1500 \times 600 \text{ kbps} = 900000$$

$$= 900 \text{ mbps}$$

50) Area covered = (πR^2)

$$= \pi \times 99.7 \times 4^2 = 50.27 \quad (\because \text{blanket } 350)$$

$$\Rightarrow \frac{250}{50.27} = 6.977$$

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blanket 400 sq km, 400m radius
cell area as R^2 , how many cells?

$$R^2 = (0.4)^2 = 0.16$$

$$\text{cells required} = \frac{400}{0.16} = 2,500$$

22 FDMA system 30 kHz, total spectrum = 6000 kHz

$$\text{channels carried} = \frac{30}{6000} = 5 \times 10^{-3}$$

23 1200 subscribers, 4 calls per hour, 2.5 min/call

$$\text{total traffic} = 1200 \times 4 \times 2.5 = 12,000$$

24 At 9.5 GHz, 40 MHz, used channels 40 kHz

$$\text{maximum number} = 9.5 \times 40 \times 1000 = 4,000,000$$

25 paging 4 cells, base station 600, 1200 each station (network comprised)

$$\text{capacity} = \frac{600}{1200} = 0.5$$

26 sweeps through 4 cells, 200-calls

$$\text{paging scans} = 4 \times 200 = 800$$

27 1,000,000 simultaneous, receives 40,000, lasting 3 minutes

$$\text{traffic end} = 40000 \times 3 = 120,000$$

28 priority left 15%, 600,000 subscribers

$$\text{Total subscribers} = 15 \times 600,000 = 90,000$$

29 capable processing 12,000, Peak load 9,000

$$\text{capacity} = \frac{\text{Peak load}}{\text{capable load}} = \frac{9,000}{12,000} = 0.75$$

30 frequency 4, 48 MHz spectrum, 200 kHz per channel

$$\text{channels} = 4 \times 48 \times 200 = 38,400$$

31 GSM deployment 200 kHz, 30 MHz allocation

$$\Rightarrow \frac{30000}{200} = 150$$

32 network logs 2500, 99% which served

$$\text{Total call} = 99\% \times 2500 = 2475$$

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⑩ vehicle travelling at 100 km/h, 900 MHz

$$100 \text{ km/h} = 27.78 \text{ m/s}$$

$$\text{Doppler shift} = v \times \frac{f}{c} = 27.78 \times \frac{900000}{3 \times 10^8} = 0.081$$

⑪ TDMA frame lasting 5 ms, info & equal free slots

$$\text{single slot} = \frac{\text{frame}}{\text{free slots}} = \frac{5}{8} = 0.625$$

⑫ 24 MHz allocated bandwidth, 900 kHz for a GSM slot

$$\text{carrier frequencies} = \frac{\text{bandwidth}}{\text{carrier spacing}} = \frac{24000}{200} = 120$$

⑬ transmits at 8W, signal experiences 33 dB

$$10 \log_{10}(8000) = 39$$

$$\text{base station} = 39 - 33 = 6 \text{ dB}$$

⑭ 1 m as the reference distance, 1900 MHz, spaced 2.5 km

$$\text{free space path} = 1 \times 1900 \times 2.5 = 4750$$

⑮ cluster consists 6 cells, 250 simultaneous calls

$$\text{total call handling} = 250 \times 6 = 1500$$

⑯ A switching subsystem 4000 new, last 3 minutes

$$\Rightarrow 4000 \times 180 \text{ s} = \frac{7,20,000}{3600} = 200$$

⑰ 150 channels to a base station, each channel 25 kHz

$$\begin{aligned} \text{Total bandwidth} &= 150 \times 25 \\ &= 3750 \\ &= 3.75 \text{ MHz} \end{aligned}$$

⑱ spectrum planners 40 MHz each 30 adjacent frequencies

$$= 40 \times 29 = 1160$$

⑲ A vehicle covers at 90 km/h, region, 1.5 km

$$\Rightarrow 1.5 \times 2 = 3000, \quad 90 \text{ km/h} = 25 \text{ m/s}$$

$$= \frac{3000 \times 25}{3600} = 20.8$$

⑳ 8 regional centers, 40 base station, each base serving 250

$$\text{total users} = 8 \times 40 \times 250 = 80,000$$

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